

PRANJAL KARAN

Computer Science Engineering Student · Full-Stack Development & AI Integration

+91-9162084158 | pranjalkaran2004@gmail.com | pranjalk.vercel.app | linkedin.com/in/pranjalk-karan | github.com/Pranjal-k22

PROFESSIONAL SUMMARY

Computer Science undergraduate (CGPA 8.70/10, Expected May 2027) with hands-on experience architecting and shipping **full-stack production platforms** using **React**, **Node.js/Express**, and **MongoDB** – including a 40,000+ line healthcare appointment system with 61 REST endpoints and database-level concurrency guarantees. Skilled at building real-time systems with **WebSockets/Socket.io**, integrating **LLM APIs** (Gemini, Ollama) into production AI pipelines, and writing automated test suites to ensure reliability. Microsoft Certified in Azure AI Cloud Development and Azure Data Fundamentals, with strong fundamentals in data structures, algorithms, and system design.

EDUCATION

VIT Bhopal University – Bachelor of Technology, Computer Science & Engineering | Sehore, MP

Sept 2023 – May 2027

CGPA: 8.70/10

TECHNICAL SKILLS

Languages	JavaScript (ES6+), C++, TypeScript
Frontend	React (18/19), TypeScript, HTML5, CSS3, Tailwind CSS v4, Zustand, Vite, Responsive Web Design
Backend & APIs	Node.js, Express.js, RESTful API Design, WebSockets, JWT Authentication, Clerk Auth
Databases	MongoDB (Mongoose, Aggregation Pipelines, Compound & Partial Unique Indexes), MySQL, PostgreSQL, SQL
Cloud & DevOps	Git, GitHub, Docker, Vercel, Vercel Serverless Functions
AI & Practices	Google Gemini, Ollama, LLM API Integration, GitHub Copilot, Agile / SDLC, Automated Testing

PROJECTS

HealthPulse – Full-Stack Healthcare Appointment & Clinical Platform | [Live](#) | [GitHub](#)

2026

React 18, TypeScript, Node.js, Express.js, MongoDB, JWT, Google Gemini, Ollama

- **Architected** a **40,000+ line** full-stack platform with **61 REST API endpoints** across 9 modules and **13 MongoDB schemas**, following a strict controller-service-model SDLC architecture.
- **Eliminated** double-booking race conditions under concurrent load by combining a **database-level compound partial unique index** with **TTL-based 5-minute slot holds**, guaranteeing atomic data integrity at the database engine tier.
- **Engineered** stateless JWT authentication with a `tokenVersion` mechanism enabling instant session revocation, role-based access control across 3 personas, and rate-limited endpoints to prevent brute-force attacks.
- **Orchestrated** a dual-engine AI clinical-summary pipeline (Google Gemini + Ollama via `Promise.allSettled`) with a zero-hallucination medication-verification guard, backed by **154 automated tests (471 assertions)** and a **98/100** Lighthouse performance score.

VibeChat – Real-Time Full-Stack Messaging Platform | [GitHub](#)

2026

React 19, Tailwind v4, Zustand, Node.js, MongoDB, Socket.io, Clerk, ImageKit, Docker

- **Engineered** a real-time messaging application achieving **sub-100ms** message distribution using custom Socket.io server channels, replacing N+1 queries with a **6-stage MongoDB aggregation pipeline** for query optimization.
- **Packaged** a lightweight **1.17MB** production bundle via Vite 8 featuring Clerk OAuth session gates and ImageKit lazy-loading pipelines.

AstroSync – AI-Powered Compatibility Analysis Platform | [Live](#) | [GitHub](#)

2026

React, TypeScript, Vite, Vercel Serverless Functions, Google Gemini API

- **Built** a TypeScript single-page application with 10 algorithmic scoring modules and interactive Recharts-based visualizations for compatibility analysis.
- **Developed** 2 Vercel serverless functions integrating the Google Gemini API with multi-model fallback and chat-history normalization to handle context-window and rate-limit constraints.

ACHIEVEMENTS & CERTIFICATIONS

Achievements **National Qualifier** – Cleared Round 1 of **TCS CodeVita**, a national-level competitive coding challenge

Certifications **Microsoft Certified: Azure AI Cloud Developer Associate** (Score: 922/1000) | **Microsoft Certified: Azure Data Fundamentals** (Score: 955/1000) | **MERN Full Stack Certification** (Ethnus & Codemithra)

LEADERSHIP

Co-Lead (Social Media) – Linpack Club, VIT Bhopal

2026

- **Directed** social media strategy and content planning for the club, growing outreach and driving a **30%** increase in event registrations through targeted digital campaigns.
- **Co-organized** MatLabVerse, coordinating promotion, scheduling, and cross-team communication to deliver a smooth participant experience.
- **Collaborated** with club leadership and peer teams to align messaging across events, strengthening the club's overall digital presence.